

Выполним эквивалентное преобразование схемы и определим комплексные параметры цепи

$$\omega = 2\pi f = 2\pi \cdot 500 = 3141.5927 \text{ rad/s}$$

$$E_1 = \frac{E_{m1}}{\sqrt{2}} e^{j\varphi_1} = 106.066 e^{j78^\circ} \text{ V}$$

$$I_1 = \frac{I_{m1}}{\sqrt{2}} e^{j\varphi_1} = 56.569 e^{j60^\circ} \text{ A}$$

$$E_2 = \frac{I_1}{G_{m2}} = \frac{28.28 + j48.99}{0.3} = 94.28 + j163.3 \text{ V}, R_{m2} = \frac{1}{G_{m2}} = \frac{1}{0.3} = 3.333 \Omega$$

$$E_{eq} = E_1 - E_2 = 93.65 + j49.79 - (94.28 + j163.3) = -0.63 - j113.5 \text{ V}$$

$$Z_{eq, n} = (R_{m1} + R_{m2}) + jX_{Lm1} = (5 + 3.333) + j2 = 8.33 + j2 \Omega = 8.57 e^{j13.5^\circ} \Omega$$

Рассчитаем эквивалентные сопротивления ветвей приемника

$$X_{L1} = \omega L_1 = 3141.5927 \cdot 0.0012 = 3.77 \Omega, X_{C1} = \frac{1}{\omega C_1} = \frac{1}{3141.5927 \cdot 0.00005} = 6366 \Omega$$

$$X_{L2} = \omega L_2 = 3141.5927 \cdot 0 = 0 \Omega, X_{C2} = \frac{1}{\omega C_2} = \frac{1}{3141.5927 \cdot 0.00003} = 1061 \Omega$$

$$X_{L3} = \omega L_3 = 3141.5927 \cdot 0.025 = 78.54 \Omega, X_{C3} = \frac{1}{\omega C_3} = \frac{1}{3141.5927 \cdot 0.000005} = 63662 \Omega$$

$$Z \quad R \quad \vee \quad \chi \quad 6 \quad 3.77 \quad 6366 \quad 6 \quad 26$$